Methodology, results, decision, and next steps.

AKIB AHMED, 24th February, 2026

Methodology

Method Applied & Reasoning

I used the Z-test for statistical significance testing. The reason i choose Z-test:

- The Z-test is specifically designed to handle binary "Success/Total" data, which matches the **Sessions-to-Clicks** and **Sessions-to-Orders** structure of this experiment.
- The Z-score provides a clear measure of how many standard deviations separate the two groups, making the results easier to interpret for **Business Intelligence** stakeholders.
- For an A/B test involving exactly two variants, the Z-test is the industry-standard choice over Chi-Square, which is typically reserved for comparing three or more groups.
- With a large sample size of over **30,000 sessions per group**, the Z-test is mathematically robust and provides a precise **P-value** for decision-making.

Data Aggregation (Summing the Successes)

Instead of looking at a single row of data, we used a Summing Methodology to capture every visitor across the entire duration of your experiment:

- ☐ **Total Successes (x_1, x_2)**: I summed all "Clicks" and "Orders" for both Group A and Group B.
- ☐ **Total Observations (n_1, n_2)**: I summed the total "Sessions" for each group to get the full sample size.

Calculating the Proportions ($\acute{p}_1$, $\acute{p}_2$)

Calculated the conversion rates for each group to see the raw performance difference:

- **Group A Rate ($\acute{p}_1$):** Total successes divided by total sessions for the Control.
- **Group B Rate ($\acute{p}_2$):** Total successes divided by total sessions for the Variant.

Pooled Sample Proportion ($\dot{P}$)

To find the **Z-score**, the formula first calculates a "Pooled Proportion," which is the weighted average conversion rate of both groups combined.

- This serves as the "Baseline" to measure how much the groups deviate from each other.
- **Formula:** $\dot{P} = (x_1 + x_2) / (n_1 + n_2)$

Computing the Z-Score

The final Z-score was calculated by taking the difference between the two proportions and dividing it by the **Standard Error** (the denominator in our formula).

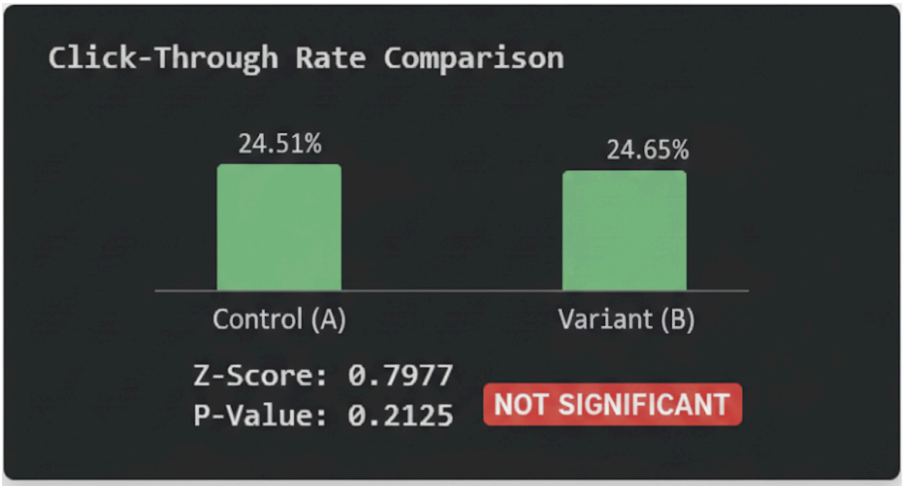
- **The Logic:** This tells us exactly how many "Standard Deviations" apart Version A and B are.
- For Click-Through Rate, this resulted in a **Z-Score of 0.7977**, which is well below the **1.96** required for significance.

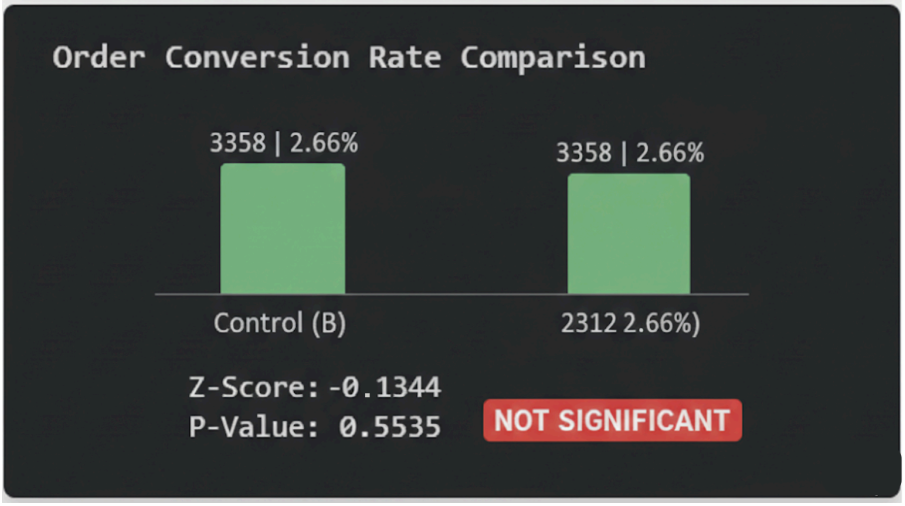
P-Value Mapping and Significance

Finally, the script mapped that Z-score to a **P-value** to determine the probability of "luck":

- **The Alpha (α):** We set a threshold of **0.05** (95% confidence).
- **The Verdict:** Because your P-values (0.2125 and 0.5535) were greater than 0.05, the methodology dictated that the results be labeled as **Neutral/Inconclusive**.

Statistical Interpretation





1. Click-Through Rate (CTR)

- **Result:** A (24.51%) vs. B (24.65%).
- **Interpretation**: While Variation B has a slightly higher raw percentage, the P-Value of 0.2125 is far above the required 0.05 threshold.
- **Meaning**: There is a 21.25% probability that this small "win" is just a random fluke. The Z-Score of 0.7977 confirms the two groups are not distinct enough to be meaningful**.**

2. Order Conversion Rate

- **Result**: Both are locked at 2.66%.
- **Interpretation**: The Z-Score of -0.1344 is effectively zero.
- **Meaning**: There is absolutely no difference in how many people are actually buying. Variation B had no impact on the bottom line.

Decision

Action	Decision	Reasoning
Results are inconclusive?	**Yes**	Click-Through Rate and Order Conversion Rate confirms the two groups are not distinct enough to be meaningful.
Roll out B?	**REJECT**	There is no proven revenue lift to justify the technical effort of a permanent rollout.
Run further?	**NO**	With over **30,000 sessions** per group, our sample size is huge. Running the test longer won't change the fact that the effect is tiny.

Proposed Experiments For Next Iteration

Option 1: The "Financial Risk-Reduction" Experiment

The goal is to address the high 910.67 AOV by lowering the psychological barrier to purchase.

- The Change: Add a **"Price Match Guarantee"** and **"Free 30-Day Returns"** badge directly under the primary CTA (Add to Cart).
- Microcopy: Change the CTA from a generic **"Add to Cart"** to **"Secure My Copy — Ships Free"**.
- Why it works: High-ticket items like those in your bookstore require "reassurance". By making the risk of a bad purchase feel like zero, you target the conversion leak at the bottom of the funnel.

Option 2: The "Comparison & Social Proof" Mix

This aligns with your long-term Research/BI goals by testing how users process information.

- The Change: Replace the static product description with a "How we compare" table and a "Verified Buyer" quote directly next to the CTA.
- Microcopy: Use **"social proof"** language: **"Join 5,000+ Readers — Add to Cart".**
- Why it works: Users often leave a page to "do more research." By providing the comparison and the review on the same page, you keep them in the funnel and increase the Order Conversion Rate, which stayed flat at 2.66% in your last test.

END